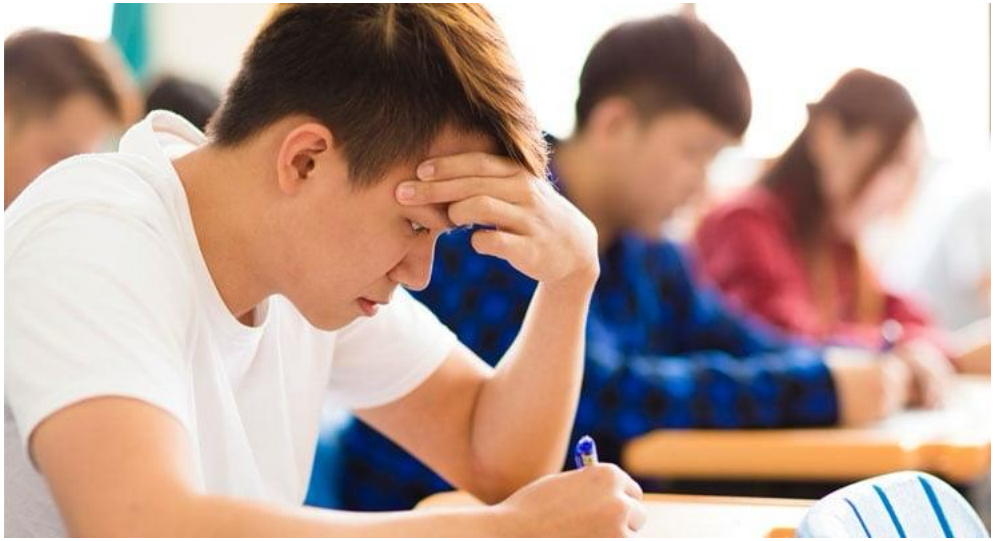
Psychometrics and the Importance of Psychological Measurement

PSY 317
Psychological Measurement

One Score. Big Consequences.



Two students:

Ayşe: 84

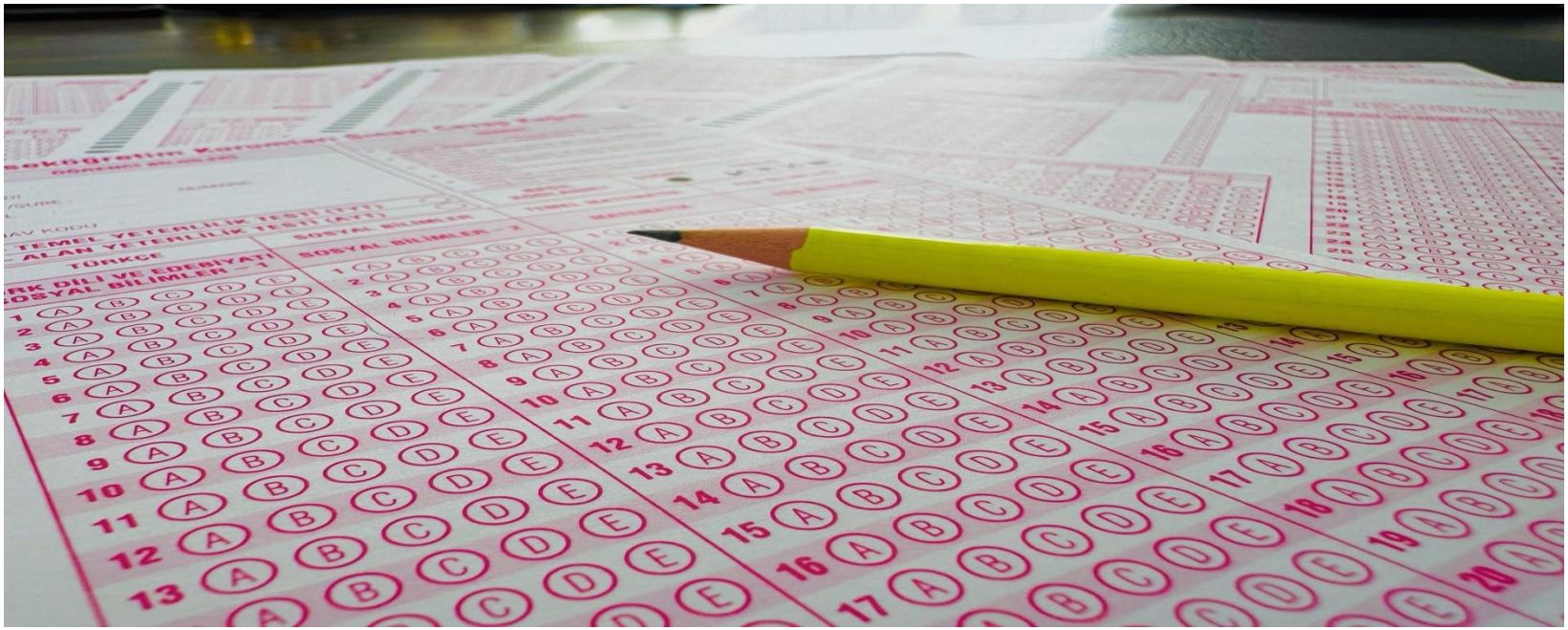
Mehmet: 86

Admission cutoff: 85

Only Mehmet is admitted.

Question:

Are they meaningfully different?



The Hidden Question:

What if:

The test contains measurement error?

The reliability is .70?

The difference is random fluctuation?

How confident should we be in decisions based on test scores?

If testing has serious consequences,

we must evaluate how strong our measurement tools are.

The Central Problem of Psychology

We cannot directly observe:

- Intelligence
- Memory
- Anxiety
- Self-esteem
- Executive functioning

- We only observe behavior.
- Psychological measurement requires **inference**. OR
- Psychological measurement involves drawing conclusions about unobservable traits based on observable behavior.

The Central Problem of Psychology

We cannot “see” working memory

Instead, we:

- Give a memory task
- Measure performance
- Infer working memory

This inference — linking behavior to mental attributes — is the core problem of psychological measurement.

Why Study Psychological Measurement?

If psychological test scores influence important decisions, we must evaluate their quality.

This course examines:

How to determine whether test scores are reliable

How to evaluate whether interpretations are valid

How to assess the precision and meaning of scores

What makes a psychological test scientifically sound

What is a Psychological Test?

Lee Joseph Cronbach
1960

A systematic procedure for comparing the behavior of two or more people. — Lee Cronbach (1960)

Key characteristics:

- Involves behavioral samples
- Collected in a standardized way
- Used to compare individuals
- Can assess:
  - Inter-individual differences
  - Intra-individual differences

Inter-individual differences: Variability between persons on a measured attribute.

Intra-individual differences: Variability within a person across time, contexts, or conditions.

Types of Test

Educational Testing Service (2011): There are tens of thousands of psychological tests in the public domain

Tests can vary "in content":

Achievement Tests	Measure a person's degree of learning, success, or accomplishment in a subject or task
Aptitude Tests	Measure the capability for a relatively specific task or type of skill
Intelligence Tests	Measure an individual's ability in relatively global areas such as verbal comprehension, perceptual organization, or reasoning
Personality Tests	Measure the traits, qualities, or behaviors that determine a person's individuallity
Neuropsychological Tests	Measure cognitive, sensory, perceptual, and motor performance to determine the extent, locus, and behavioral consequences of brain damage

Types of Test

Tests also vary "with regard to the type of response required":

Open-Ended Tests	People can answer test questions by saying anything they want in response to the questions on the test
Closed-Ended Tests	Require people to answer questions by choosing among alternative answers provided in the test

Tests also vary "according to the methods used to administer them":

There are individually administered tests, and there are tests designed to be administered to groups of people

Speeded Tests	Time-limited and scored by counting the number of questions answered in the allotted time
Power Tests	Not time-limited and scored by counting the number of correct answer made on the test

Types of Test

Another common distinction "concerns the intended purpose of test scores":

Criterion Referenced Tests	Seen in settings in which a decision must be made about a person's skill level A fixed, *predetermined cutoff test score* is established, and it is used to sort people into two groups: 1. Those whose performance exceeds the performance criterion 2. Those whose performance does not
Norm Referenced Tests	Used to compare a person's test score with scores from a *reference sample* or a *normative sample*, in order to understand how the person compares with other people Characteristics of the *reference sample* are thought to be representative of some well-defined population

What is Psychometrics?

Just as;

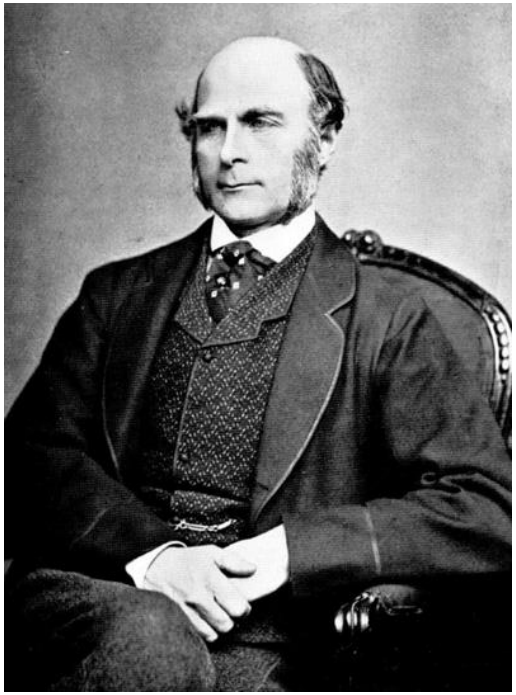
- Psychological tests are designed to measure psychological attributes of people, psychometrics is the science concerned with evaluating the attributes of psychological tests
- Psychological attributes of people must be measured, so also psychometric attributes of tests must be estimated

Psychometrics is about the procedures used to estimate and evaluate the attributes of tests

Three of these attributes will be of particular interest:

1. The *type of information* generated by the use of psychological tests
2. The *reliability* of data from psychological tests
3. Issues concerning the *validity* of data obtained from psychological tests

Historical Roots



Francis Galton (1822-1911)

Francis Galton introduced early measurement of human characteristics.

He referred to the measurement of mental processes as:

"The art of imposing measurement and number upon operations of the mind."

His work laid important foundations for psychometrics, though some ideas later became associated with controversial movements such as eugenics.

Challenges to Measurement in Psychology

The complexity of psychological phenomena

Try to identify and capture the important aspects of these types of human psychological attributes in a single number

Participant reactivity

Psychologists are measuring psychological characteristics of people who are conscious and generally know that they are being measured, the act of measurement can itself influence the psychological state or process being measured

- *Demand characteristics*: Changing behavior to accommodate researcher
- *Social desirability*: Changing behavior to impress the person doing the measurement
- *Malingering*: Changing behavior to convey a poor impression the person doing the measurement

All Measurement Contains Error

Every observed score consists of:

Observed Score = True Score + Error

Error may come from:

- Temporary mood
- Fatigue
- Guessing
- Ambiguous items
- Environmental distractions

Psychometrics does not eliminate error — it estimates and manages it.

Challenges to Measurement in Psychology

Expectation and bias

The people collecting the behavioral data can bring biases and expectations to their task

Score sensitivity

The ability of a measure to discriminate adequately between meaningful amounts or units of the dimension that is being measured

Check the box below that best describes your general emotional state over the past week

☐ Good	☐ Bad

1	2	3	4	5	6	7	8	9
Extremely Good								Extremely Bad

Lack of awareness of important psychometric information